Evaluating the performance gap between selected tree-based methods and a modern transformed-based Tweet classifier.

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Abstract- This project benchmarks the performance gap between traditional machine learning methods and a state-of-the-art transformer model for sentiment analysis. Specifically, it compares Random Forest and XGBoost against the XLM-T model on English Twitter data from the CardiffNLP dataset. The primary objective is to assess whether traditional models can remain competitive with transformer-based approaches in this domain. The core methods involve training the machine learning models using TF-IDF vectorization and fine-tuning the pre-trained XLM-T model. Model performance is evaluated using the macro-F1 score, accuracy precision and recall. The key findings confirm a significant performance gap, with the fine-tuned XLM-T model substantially outperforming the traditional tree-based methods, supporting the hypothesis that modern transformer architectures are superior for this task.

I. INTRODUCTION

Sentiment analysis, the task of automatically determining the emotional polarity of text, is crucial for understanding public opinion, user feedback, and social discourse (Nandwani and Verma, 2021). Common approaches historically relied on rule-based systems or classical machine learning models operating on handcrafted features (Alslaity and Orji, 2024). However, the advent of deep learning, particularly transformer architectures, has revolutionized Natural Language Processing (NLP), achieving state-of-the-art results on various tasks, including sentiment analysis. This project aims to benchmark traditional machine learning models against a transformer-based model. This topic was chosen given that the dataset is well-curated, robust, and well-suited for sentiment analysis. Our interest in the platform formerly known as Twitter (now "X") further motivates this work. This has significant relevance in business, social, and policy contexts, including applications in market research, brand monitoring, hate speech detection, mental health monitoring, and public opinion analysis.

Particularly, this work aims to compare the performance on sentiment analysis of English Twitter data of Random Forest (RF) and XGBoost (XGB) against XLM-T, a SOTA RoBERTa based model (Barbieri, et al. 2022). For this matter, the macro-F1 score is used to assess performance across all sentiment classes equally. In that sense, we anticipate a significant performance gap favoring the transformer model, given that classic machine learning techniques will not achieve the same level of performance as modern state-of-the-art approaches like transformers, specifically in NLP tasks such as sentiment analysis (Patwardhan, et al. 2023).

II. RELATED STUDY

The landscape of sentiment analysis has evolved significantly, with various machine learning and deep learning techniques being applied to this task

A systematic review examining 123 studies on machine learning for emotion detection was conducted by Alslaity and Orji (2024). Their findings indicated that supervised machine learning algorithms, specifically Support Vector Machines (SVM) and Naïve Bayes, are the most frequently employed methods, alongside tree-based algorithms. The study also revealed that English-language text datasets are the predominant data source in this field, and accuracy is the most commonly used metric for performance evaluation. Rathi et al. (2018) focused on enhancing classification outcomes in sentiment analysis by employing ensemble machine learning techniques to boost the efficiency and reliability of their approach; their specific implementation combined a Support Vector Machine with a Decision Tree. In the context of sentiment analysis on Twitter data, Jain and Dandannavar (2016) utilized Naïve Bayes and Decision Tree algorithms, testing them on datasets of varying sizes (200, 2000, and 4000 tweets). Their results demonstrated that the Decision Tree algorithm required significantly less time to predict unseen data compared to Multinomial Naïve Bayes and generally outperformed it across the different dataset sizes and domains considered. Neelakandan and Paulraj (2020) applied a Gradient Boosted Decision Tree (GBDT) classifier for sentiment classification of Twitter data, incorporating preprocessing steps like tokenization, stop-word removal, filtering, lemmatization, and hashtag handling. They compared the GBDT's performance against various deep neural network models using metrics including precision, recall, F-score, execution time, and accuracy, concluding that GBDT delivered superior results relative to the compared deep neural network models. Hama Aziz and Dimililer (2021) developed an ensemble classifier named SentiXGboost, which integrated multiple feature sets (such as Bag of Words, TF-IDF, Part of Speech, N-gram,

Opinion Lexicon, and Term Frequency) through ensemble classification. The outputs from individual classifiers trained on these features were fed into an XGBoost meta-classifier, and the system was evaluated using accuracy, precision, recall, and F-score.

However, the field has seen a significant shift towards deep learning models, particularly transformer architectures. Kokab et al. (2022) investigated efficient classification techniques using transformer models, comparing the performance of BERT, RoBERTa, ALBERT, and DistillBERT against Long Short-Term Memory (LSTM) networks. Their findings showed RoBERTa achieved the highest performance with 95% accuracy, followed by BERT at 93%, while ALBERT and DistillBERT achieved accuracies closer to BERT, although ALBERT demonstrated a lower processing time. This trend is reinforced by recent work: Albladi et al. (2025) proposed TWSSenti, a hybrid framework combining BERT, GPT-2, RoBERTa, XLNet, and DistilBERT, achieving up to 95% accuracy on Sentiment140 and IMDb datasets. Similarly, Semary et al. (2023) integrated RoBERTa embeddings with CNN and LSTM layers, reporting 96.28% accuracy on IMDb and 94.2% on Twitter data. Bashiri and Naderi (2024) conducted a large-scale comparative review of eight transformer models, finding T5 to be the most consistent performer, while RoBERTa and ELECTRA excelled on specific datasets. Joshy et al. (2022) evaluated BERT, DistilBERT, and RoBERTa on Sentiment140 and COVID-19 tweets, confirming BERT's superior accuracy despite efficiency trade-offs. Habib et al. (2024) explored multimodal sentiment analysis by fusing text and image data, where RoBERTa combined with EfficientNet achieved promising results. Finally, Obiorah et al. (2025) compared lexicon-based, traditional machine learning, and transformer models, concluding that DistilBERT-SST-2 and RoBERTa-Sentiment significantly outperformed older approaches in accuracy and F1-score.

III. METHODOLOGY

This study employs a structured approach to sentiment analysis, integrating both traditional machine learning techniques and transformer-based models. All procedures were implemented using Python within a Jupyter Notebook used for traditional machine learning experiments and Google Colab for deep learning models to leverage GPU acceleration. The workflow utilized libraries such as scikit-learn, XGBoost, and HuggingFace Transformers for model development, and NLTK for text preprocessing. Visualization and evaluation were supported by Seaborn, and WordCloud, ensuring clarity and reproducibility throughout the process.
The methodology encompasses five key stages: data collection, preprocessing, semantical cleaning, feature extraction, and model training and evaluation. A detailed description of these stages, along with the experimental setup and evaluation metrics, is provided in the following subsections. The overall workflow is summarized in Figure 1, which illustrates the six stages of the methodology and their interconnections.

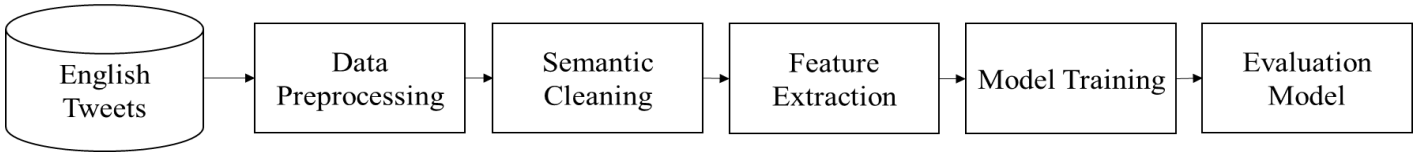

Figure 1: Flowchart of the sentiment analysis methodology, showing the six key stages from data collection to model evaluation.

1. Data description: The dataset used in this study was obtained from the HuggingFace repository, specifically the CardiffNLP Tweet Sentiment Multilingual corpus. While the original dataset includes tweets in multiple languages, this project focuses exclusively on English-language tweets to maintain consistency and simplify preprocessing. The combined dataset, created by merging the original training, validation, and test splits, consists of 3,033 records and two columns: text, which contains the tweet content, and label, which represents the sentiment class. There are three sentiment categories: 0 for negative, 1 for neutral, and 2 for positive. The distribution of sentiment classes is perfectly balanced, with 1,011 negative tweets (33.3%), 1,011 neutral tweets (33.3%), and 1,011 positive tweets (33.3%). Despite this balance, the relatively small dataset size may limit generalization, and since all data originates from Twitter, platform bias is a potential limitation. Additionally, tweets often include informal language, abbreviations, and emojis, introducing noise that requires careful preprocessing.

2. Data Preprocessing: Data preprocessing ensures raw tweets are transformed into a clean and structured format suitable for analysis. The dataset was checked for missing values and duplicates, which were removed to maintain integrity. Text normalization was applied by converting all text to lowercase and removing URLs, special characters, numeric sequences, and extra whitespace using regular expressions. These steps were implemented in Python using libraries such as **pandas** for data manipulation and re for regular expressions.

3. Semantical Cleaning: Semantic cleaning focused on improving linguistic quality. Stop words, mentions, and hashtags were removed while retaining meaningful tokens. Lemmatization using NLTK's WordNetLemmatizer reduced words to their base form, and tokenization prepared the text for feature extraction. Word clouds were generated to visualize raw and cleaned text distributions, providing an intuitive overview of the preprocessing impact.

4. Feature Extraction: Feature extraction was performed using TF-IDF vectorization for traditional machine learning models, converting text into numerical feature vectors. For the transformer-based model (XLM-T), tokenization and padding were implemented using HuggingFace's AutoTokenizer to prepare inputs for fine-tuning.

5. Model Training and Testing: The dataset was split into training (70%), validation (15%), and testing (15%) subsets. Random Forest and XGBoost models were trained using TF-IDF features, while XLM-T was fine-tuned using HuggingFace's Trainer API. All experiments were conducted in Python using scikit-learn, XGBoost, and Transformers libraries.

6. Evaluation Metrics: Model performance was assessed using Accuracy, Precision, Recall, and Macro-F1 Score to ensure balanced evaluation across all sentiment classes. Confusion matrices were also generated for visual performance analysis.

IV. Experimental results

The experimental results comparing the performance of Random Forest, XGBoost, and fine-tuned XLM-T on the sentiment analysis task are presented below. This section includes a summary of key metrics and confusion matrices to illustrate classification performance across the three sentiment classes.

Performance summary

The Random Forest classifier achieved an overall accuracy of 0.51. Its performance varied across the three sentiment classes: it showed the highest F1-score for the 'neutral' class at 0.54, followed by 'negative' at 0.49, and 'positive' at 0.51. Notably, for the 'positive' class, Random Forest demonstrated the highest recall (0.64) compared to XGBoost (0.59) and XLM-T (0.47), meaning it correctly identified more positive instances than the other two models. The macro-F1 score was 0.51. The XGBoost classifier also achieved an overall accuracy of 0.51. Its F1-scores per class were 0.50 for 'negative', 0.49 for 'positive' and 0.55 for 'neutral'. The macro-F1 score was 0.52. The fine-tuned XLM-T model demonstrated significantly superior performance, achieving an overall accuracy of 0.71. Its F1-scores were notably higher across all classes compared to the traditional models: 0.795 for 'negative', 0.554 for 'positive' and 0.754 for 'neutral'. The macro-F1 score was 0.70.

Table 1: Precision across the three evaluated models

Metric	RF Precision	XGB Precision	XLM-T Precision
Class 0	0.62	0.57	0.75
Class 1	0.42	0.41	0.68
Class 2	0.58	0.63	0.69
Macro Avg	0.54	0.54	**0.71**
Weighted Avg	0.54	0.54	0.71

Table 2: Recall across the three evaluated models

Metric	RF Recall	XGB Recall	XLM-T Recall
Class 0	0.4	0.45	0.84
Class 1	0.64	0.59	0.47
Class 2	0.5	0.49	0.83
Macro Avg	0.51	0.51	**0.71**
Weighted Avg	0.51	0.51	0.71

Table 3: F1-Score across the three evaluated models

Metric	RF F1	XGB F1	XLM-T F1
Class 0	0.49	0.5	0.80
Class 1	0.51	0.49	0.55

Class 2	0.54	0.55	0.75
Macro Avg	0.51	0.52	**0.70**
Weighted Avg	0.51	0.52	0.70

CONFUSION MATRICES

Figures 2, 3, and 4 present the confusion matrices for Random Forest, XGBoost, and XLM-T, respectively. These matrices illustrate the distribution of correct and incorrect predictions across the three sentiment classes (negative, neutral, positive). They complement the quantitative metrics by showing where misclassifications occur.

The confusion matrix for Random Forest reveals strong performance on the neutral class (94 correct predictions). However, there is significant confusion between neutral and positive (59 positive tweets predicted as neutral) and between negative and neutral (71 negative tweets predicted as neutral). Despite these misclassifications, Random Forest achieved the highest recall for positive (0.64), reflected in the relatively high number of correctly predicted positive tweets (75).

XGBoost performs similarly to Random Forest, with its confusion matrix showing frequent misclassification between neutral and positive (56 positive tweets predicted as neutral) and negative tweets often predicted as neutral (69 cases). It slightly improves negative class predictions (71 vs. 63 for Random Forest) but still struggles to distinguish neutral and positive sentiments.

XLM-T demonstrates clear improvements across all classes. Negative tweets are predicted correctly in most cases (132), and positive tweets have the highest correct predictions (124), which aligns with its superior F1-scores. Some confusion remains between neutral and positive (44 positive tweets predicted as neutral), but it is significantly lower than in traditional models. The balanced distribution of correct predictions explains XLM-T's overall accuracy of 0.71 and macro-F1 of 0.70.

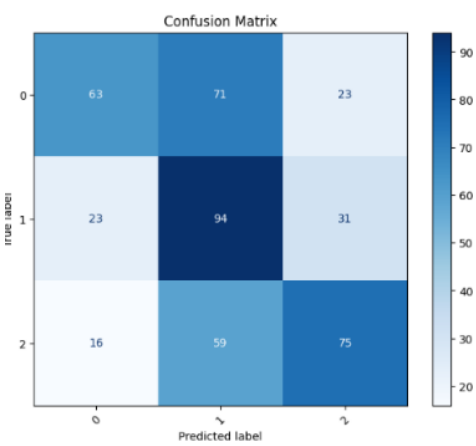

Figure 2 Confusion Matrix Random Forest

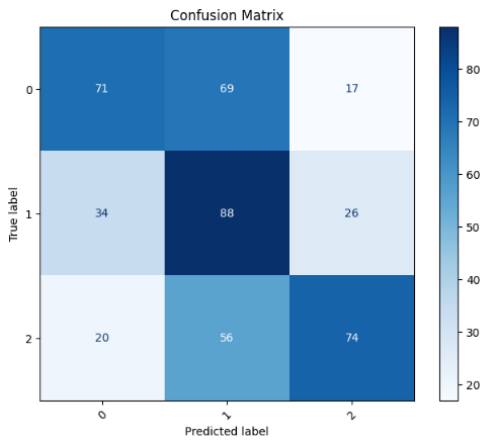

Figure 3 Confusion Matrix XGBoost

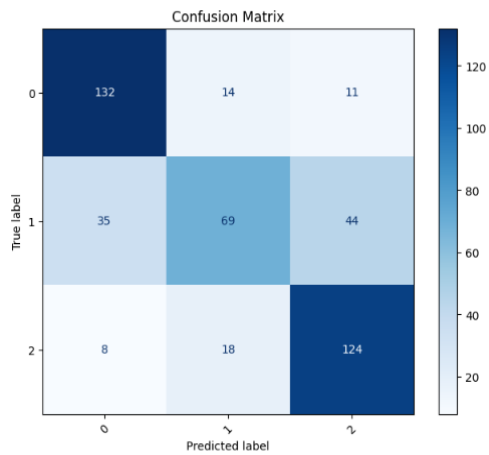

Figure 4 Confusion Matrix XLM-T

V. CONCLUSION

This project benchmarked the performance of Random Forest and XGBoost against the state-of-the-art transformer model XLM-T (Barbieri, et al. 2022) for sentiment analysis on English Twitter data. The findings confirm a significant performance gap, with the fine-tuned XLM-T model substantially outperforming the traditional models in terms of overall accuracy and macro-F1 score. While Random Forest showed a specific strength in recall for the 'positive' class, the overall results support the hypothesis that transformer-based models are superior for this NLP task.
The confusion matrices provide additional insight into these differences: traditional models exhibited frequent misclassifications between neutral and positive tweets, as well as between negative and neutral, indicating challenges in distinguishing subtle sentiment variations. In contrast, XLM-T demonstrated a more balanced distribution of correct predictions across all classes, reducing these confusions and reinforcing its superior performance.
These results highlight the trade-off between model complexity and performance, suggesting that for high-accuracy sentiment analysis on social media text, the computational cost of transformer models may be justified.

Based on the findings and the initial proposal, several avenues for future work are suggested:

Preprocessing Alignment: As noted in the methodology review, applying the same preprocessing steps (like lemmatization and URL/mention removal) used for traditional models directly to the transformer input might be suboptimal.

- Advanced Vectorization for ML: Explore more sophisticated text vectorization techniques for the traditional machine learning models beyond TF-IDF. Investigating methods like Word2Vec, GloVe, or even simple pre-trained embeddings could potentially improve their performance.
- Ensemble Methods: Future work could involve creating ensembles not just of base learners like in XGBoost, but potentially combining predictions from the best-performing traditional model and the transformer model to see if it further improves performance.

AUTHOR NOTE

The authors used OpenAI's ChatGPT and Qwen to assist with wording improvements and minor edits for clarity. All methodological decisions, analyses, and conclusions were made entirely by the authors.

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